Project 1: Dynamic Programming

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Submission

Please submit the following files on **Gradescope** by the deadline shown at the top right corner.

1. **Programming assignment**: upload all code you have written for the project and a README file with a clear, concise description of the main file and how to run your code.

2. **Report**: upload your report in pdf format. You are encouraged but not required to use an IEEE conference template[1] for your report. Please refer to Slide 8 of Lecture 1 for the expected report structure and contents[2].

Project Description

This project focuses on autonomous navigation in a **Door & Key** environment, shown in Fig. 1. The objective is to get our agent (red triangle) to the goal location (green square). The environment may contain a door which blocks the way to the goal. If the door is closed, the agent needs to pick up a key to unlock the door. The agent has three regular actions, *move forward* (MF), *turn left* (TL), and *turn right* (TR), and two special actions, *pick up key* (PK) and *unlock door* (UD). Taking any of these five actions costs energy (positive cost), specified in Table 1. For example, an action sequence minimizing the cost in Fig. 1 is:

$$\begin{aligned} PK &\rightarrow MF \rightarrow MF \rightarrow TR \rightarrow UD \rightarrow MF \rightarrow MF \\ &\rightarrow TR \rightarrow MF \rightarrow MF \rightarrow MF \rightarrow TL \rightarrow MF \end{aligned}$$

Design and implement a Dynamic Programming algorithm that minimizes the cost of reaching the goal in two scenarios.

(A) *"Known Map"*: You should compute different control policies for each of the 7 environments provided in the starter code and evaluate their performance on the corresponding environment. Further description of the 7 environments and their functionality is provided in the accompanying starter code and README file.

(B) *"Random Map"*: You should compute a *single* control policy that can be used for *any* of the 36 random 10×10 environments. The size of the grid in the random maps is 10×10 and the perimeter is surrounded by walls. Denote the index of the top-left cell as $(0, 0)$, bottom-left as $(0, 9)$, top-right as $(9, 0)$ and bottom-right as $(9, 9)$. There is a vertical wall at column 5 with two doors at $(5, 3)$ and $(5, 7)$. Each door can either be open or locked (requires a key to open). The key is randomly located in one of three positions $\{(2, 2), (2, 3), (1, 6)\}$ and the goal is randomly located in one of three positions $\{(6, 1), (7, 3), (6, 6)\}$. The agent is initially spawned at $(4, 8)$ facing up.

[1] https://www.ieee.org/conferences/publishing/templates.html
[2] https://natanaso.github.io/ece276b/ref/ECE276B_1_Introduction.pdf

Table 1: Non-Uniform Action Costs

Action	**Cost**	**Logic**
Turn (TL, TR)	1	**Lowest:** Turning in place is energy-efficient; the agent should prefer rotating over moving.
Move Forward (MF)	3	**Higher:** Traversing a grid cell costs more than a simple rotation.
Pick Up Key (PK)	2	**Medium:** Picking up a key is more difficult than rotating, but less costly than moving forward.
Unlock Door (UD)	5	**Highest:** A complex physical interaction that should be the most "expensive" action.

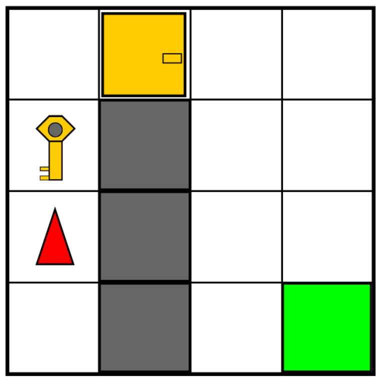

Figure 1: Door & Key Environment: An agent (red triangle) needs to navigate to a goal location (green square). The white squares are traversable, while the gray squares are obstacles that the agent cannot go through. The agent may need to pick up a key if a door along the way to the goal.

Project Report

Write a project report describing your approach to the Door & Key problem. Include the following sections.

1. [5 pts] **Introduction:** Provide a short introduction to the problem, its potential application to robotics, and your approach at a high level.
2. [25 pts] **Problem Statement:** Formulate the problem as a Markov Decision Process. Define the state space $\mathcal{X}$, control space $\mathcal{U}$, motion model f or p_f, initial state $\mathbf{x}_0$, planning horizon T, stage cost ℓ, terminal cost $\mathfrak{q}$, and any other elements necessary to make this a well-defined problem.
3. [20 pts] **Technical Approach:** Describe your algorithm for obtaining an optimal control policy that minimizes the agent's cost. This should be a clear description of the algorithm that you implemented in python.
4. [20 pts] **Results:** Use your python implementation to determine optimal control policies for the *"Known Map"* and *"Random Map"* scenarios. Your results should include visualizations of the trajectory followed by the agent with different starting positions and orientations for each environment. The results section should also include a discussion of your algorithm's performance: what worked, what did not, and why.

In addition to the report, your submission should include the code you write to solve the Door & Key problem.

5. [30 pts] **Code:** Your code will be evaluated based on *correctness* and *efficiency*. Correctness refers to whether your code implements the algorithm you present in your report and whether the algorithm you present actually solves the problem you formulated. Efficiency refers to your code being able to execute and produce control sequences for the agent in reasonable time. The results presented in the report should be consistent with the output of your code. The code should have a clear structure and comments[redacted]